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Comprehensive Review on Deep Learning and Foundation Models in Agriculture: Progress, Benchmarks, and Open Challenges

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Abstract

Deep learning has enabled automated perception, prediction, and decision support in precision agriculture, but the evidence base remains fragmented across tasks, sensing modalities, and evaluation protocols, limiting reproducible deployment. This review aimed to consolidate architectures, applications, datasets, performance metrics, and adaptation practices in agricultural deep learning from 2010 to early 2026. We conducted a structured, PRISMA-guided systematic review across major scholarly databases, screened 315 records, and synthesized 89 peer-reviewed studies using a thematic-chronological framework. We analyzed model families spanning convolutional neural networks, recurrent neural networks, encoder-decoder segmentation models, transformers, generative adversarial networks, reinforcement learning, and emerging domain-specific large language models and vision-language models, and we critically assessed validation practices, bias risks, and fine-tuning strategies under limited labeled data. The synthesis supported more comparable benchmarking and informed model selection and adaptation for reliable field-scale agricultural deployment.

Keywords

agricultural llms, and vision-language models, bioengineering, crop disease and weed analytics, deep learning architectures, precision agriculture

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***Abstract*—Deep learning has enabled automated perception, prediction, and decision support in precision agriculture, but the evidence base remains fragmented across tasks, sensing modalities, and evaluation protocols, limiting reproducible deployment. This review aimed to consolidate architectures, applications, datasets, performance metrics, and adaptation practices in agricultural deep learning from 2010 to early 2026. We conducted a structured, PRISMA-guided systematic review across major scholarly databases, screened 315 records, and synthesized 89 peer-reviewed studies using a thematic–chronological framework. We analyzed model families spanning convolutional neural networks, recurrent neural networks, encoder–decoder segmentation models, transformers, generative adversarial networks, reinforcement learning, and emerging domain-specific large language models and vision–language models, and we critically assessed validation practices, bias risks, and fine-tuning strategies under limited labeled data. The synthesis supported more comparable benchmarking and informed model selection and adaptation for reliable field-scale agricultural deployment.**

***Index Terms*—Precision agriculture, Deep learning architectures, Crop disease and weed analytics, agricultural LLMs, and vision–language models**

I. Introduction

Precision agriculture requires reliable perception and prediction to reduce yield loss, constrain chemical inputs, and enable timely field interventions. These objectives depend on extracting agronomic signals from heterogeneous sources, including proximal sensors, smartphone imagery, unmanned aerial vehicle (UAV) data, satellite observations, and farm records. Deep learning supports these requirements by learning task-specific representations directly from raw inputs and scaling to field-level inference when embedded in automated pipelines. However, the literature remains difficult to operationalize because reported performance is frequently conditioned on controlled imaging, narrow crop or regional coverage, and validation designs that permit spatial or temporal leakage. Cross-study comparison is further limited by inconsistent dataset disclosure, non-uniform metric definitions, and incomplete reporting of adaptation and deployment constraints. This evidence gap complicates architecture selection, domain transfer, and performance interpretation for real deployments.

This study synthesized and critically evaluated peer-reviewed research on agricultural deep learning from 2010 to early 2026 using a thematic-chronological structure spanning architectures, tasks, platforms, datasets, metrics, and fine-tuning. The review addressed one primary question:

- **RQ1:** Which deep learning architectures and adaptation strategies most consistently supported robust agricultural perception and prediction across tasks and data regimes from 2010 to early 2026? Two secondary questions guided critical appraisal:
- **RQ2:** How did validation protocols, dataset characteristics, and metric choices influence comparability of reported performance?
- **RQ3:** Which technical constraints most limit field-scale deployment, and which directions offer testable solutions for the next generation of agricultural deep learning systems?

Within the synthesized evidence, convolutional neural networks (CNNs) remained strong baselines for disease classification and weed segmentation [1], recurrent and hybrid networks were commonly reported for yield forecasting [2], encoder–decoder designs achieved high segmentation accuracy [3], transformer-based models were increasingly evaluated for complex weed detection [4], generative adversarial networks supported data augmentation under limited labels [5], reinforcement learning was applied to autonomous harvesting [6], and domain-specific language and vision–language models were early-stage [7]. The synthesis emphasized methodological comparability by tracking split design, metric selection, and adaptation practice across themes.

The review followed a PRISMA-guided selection protocol (Fig. 1). Searches were conducted in Google Scholar, ScienceDirect, IEEE Xplore, MDPI, Frontiers, and arXiv (Jan. 2010–Jan. 2026) using combined agriculture and model-family terms (e.g., “plant disease”, “weed”, “yield prediction” with “CNN”, “U-Net”, “transformer”, “GAN”, “large language model”, and “vision-language model”). Eligible studies were English peer-reviewed journal or conference papers that eval-

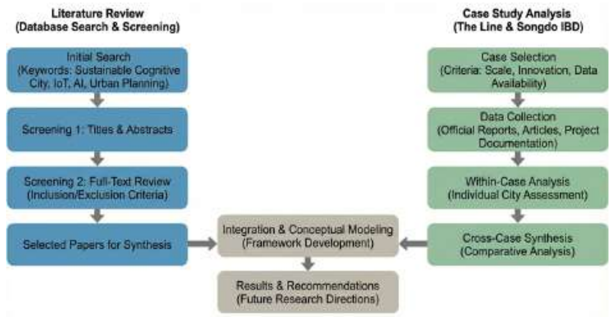

Fig. 1: Protocol for selecting articles and preparing this study.

uated a deep learning method in an agricultural context and reported explicit metrics on an identifiable dataset. The screening flow was: 315 records identified; 74 duplicates removed; 241 screened by title/abstract; 152 excluded; 89 full texts included in qualitative synthesis. For each study, extraction focused on (i) task and sensing modality, (ii) model family, (iii) dataset and split protocol (including leakage controls when reported), and (iv) evaluation and adaptation details (metrics, validation protocol, augmentation, transfer learning, fine-tuning, domain adaptation, retrieval-augmented methods). Evidence was synthesized narratively with attention to bias risks (curated datasets, leakage, incomplete threshold and uncertainty reporting); conflicting findings were interpreted through data regime, sensing modality, split design, and metric definition. The remainder of this article organizes evidence by architecture and task, then consolidates deployment platforms, benchmark datasets, evaluation metrics, and fine-tuning strategies, and concludes with research directions centered on multimodal fusion, domain adaptation, uncertainty-aware validation, and responsible field deployment.

II. Deep Learning Architectural Landscape

Deep learning has shifted the landscape of precision agriculture by enabling machines to learn complex patterns from high-volume data. Early surveys noted that neural networks outperformed classical image processing for tasks such as disease classification and yield regression. In this section, we review the major network architectures used in agricultural research from 2010 to early 2026.

A. Feedforward and Shallow Networks

Multilayer perceptrons (MLPs) or fully connected neural networks were among the first architectures applied to agronomic prediction. These models process tabular inputs such as soil properties and meteorological readings. Hybrid models often combine a shallow network with another module to capture temporal structure. For example, a soil-moisture forecasting system integrated a feedforward network with an LSTM and XGBoost regression to achieve R^2 values around 0.985 for forecasts up to one week [8]. Feedforward networks are computationally efficient but lack spatial awareness.

B. Convolutional Neural Networks (CNNs)

CNNs are the backbone of most vision-based agricultural applications. By sharing weights across local receptive fields, CNNs detect translation-invariant features and require fewer parameters than fully connected networks. Early models used architectures such as AlexNet and VGG to classify leaf images from the PlantVillage dataset and achieved accuracy above 99% [9]. Subsequent studies adopted deeper models, including ResNet and EfficientNet, and applied transfer learning from ImageNet. Lightweight variants such as MobileNet and EfficientNetB0 allow on-device inference. An edge-oriented pest detection model fused MobileNetV2 with EfficientNetB0 and used pruning and quantization; it achieved 89.5% overall accuracy with precision 95.68% and F1-score 95.67% while running on low-power devices [10].

C. Recurrent Neural Networks (RNNs)

RNNs process sequential inputs and capture temporal dependencies. Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) variants are widely adopted. A CNN-RNN hybrid predicted corn and soybean yields using historical weather and management data; it reported root-mean-square errors of 9% and 8% of average yields and generalized to unseen environments [2]. DualSpinNet combined LSTM and GRU streams with a final GRU, improving yield estimates by modeling seasonally varying factors [11]. Hybrid LSTM-XGBoost models have also been used to predict soil moisture and show that combining deep and tree-based models can capture both temporal and non-linear relationships [8].

D. U-Net and Encoder–Decoder Models

Segmentation tasks require models that provide dense predictions. U-Net employs an encoder–decoder architecture with skip connections to preserve spatial details. It has been widely used to separate crops and weeds in high-resolution images. A concatenated attention U-Net added convolutional block attention modules and achieved 99.09% accuracy, mean intersection-over-union of 81.02% and F1-score 99.06%, using only 0.377 million parameters [3]. EDM-UNet introduced efficient channel attention, Canny edge detection and morphological post-processing to reach 89.45% MIoU and 93.53% recall on UAV weed segmentation, with inference speed of 40 fps [12]. U-Net variants deliver accurate segmentation while remaining lightweight enough for field deployment.

E. Generative Adversarial Networks (GANs)

Data scarcity is a major challenge in agricultural vision. GANs generate realistic synthetic images to augment training sets and reduce overfitting. A 2022 review noted that since 2017 researchers have applied GANs to augment data for plant health assessment, weed detection, fruit quality evaluation and aquaculture [5]. GAN-based augmentation improved wheat yield segmentation by balancing class distributions and reducing domain bias [13]. Temporal GANs generate synthetic sensor time series for tasks such as irrigation scheduling. While GANs help mitigate data shortages, training can be

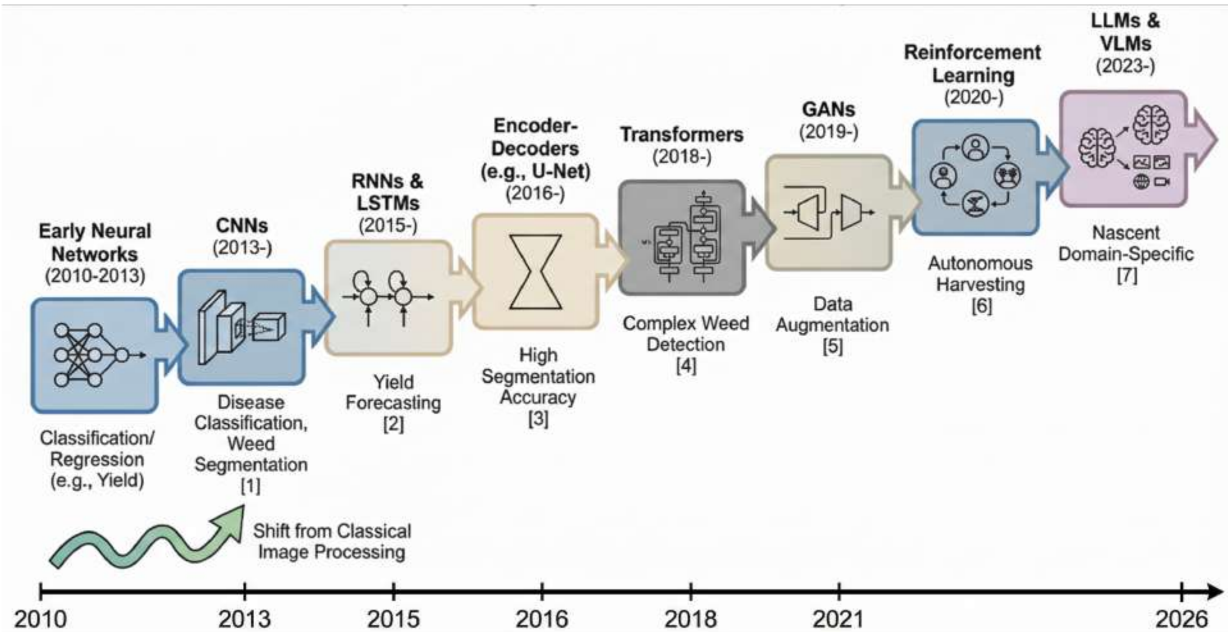

Fig. 2: Timeline of deep learning models.

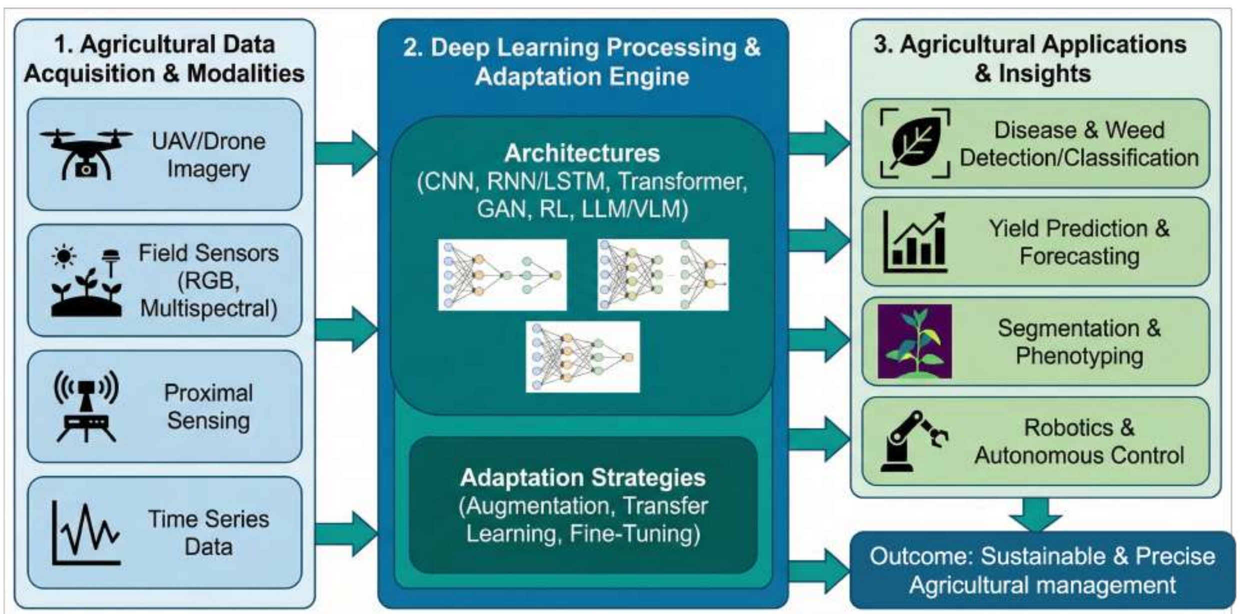

Fig. 3: Applications of deep learning in agriculture (detection, segmentation, classification and prediction).

unstable and synthetic data may not fully capture rare field conditions.

F. Object Detection Frameworks

Real-time detection of weeds, pests and fruits relies on single-stage detectors such as YOLO and two-stage detectors such as Faster R-CNN and RetinaNet. YOLO is the most widely adopted architecture for weed detection [14]. HDMS-YOLO incorporated hierarchical feature modules and multi-scale aggregation, achieving 74.2% accuracy, 66.3% recall and 71.2% mAP, outperforming a baseline YOLOv5 by 2–3 percentage points [15]. YOLO-Weed-Nano used depth-wise separable convolutions and a bi-directional feature pyramid network, reducing parameters by 63.8% and computation by 42% while improving mAP by 1% [16]. WeedSwin introduced a hierarchical transformer integrated with SAM-2 and achieved 0.993 ± 0.004 mAP and 0.985 mean average recall at 218 fps across diverse weed species and growth stages [17]. Two-stage detectors offer higher accuracy but greater computational cost; they are used mainly in research rather than field deployment.

G. Segment Anything and Vision Transformer Models

Foundation models seek broad generalization. Meta's Segment Anything Model (SAM) provides zero-shot segmentation; researchers have used SAM to generate weed and crop masks without manual labeling, subsequently training U-Nets to produce accurate weed maps [18], [19]. Vision Transformers (ViTs) model global dependencies using self-attention. WeedNet-ViT achieved 96.9% accuracy on the DeepWeeds dataset using class weighting and data augmentation [20]. WeedSwin's hierarchical transformer delivered high accuracy and real-time speed [17]. Transformers are also used in detection frameworks such as DINO and DETR; however, they require large datasets and significant computing power.

H. Large Language Models and Vision–Language Models

Large language models (LLMs) provide text generation and reasoning capabilities. Several domain-specific LLMs exist for agriculture, including AgroGPT, AgroLLM, AgriLLM and AgriBERT [7]. Many of these models lack multi-modal reasoning, prompting the development of AgriGPT, which uses a multi-agent data engine and Tri-RAG retrieval-augmented generation. AgriGPT built a 342 k question–answer dataset and outperformed general LLMs on agricultural tasks [7]. Vision–language models (VLMs) combine image encoders with language decoders. The AgroBench benchmark evaluated VLMs on 203 crop categories and 682 disease categories; most zero-shot VLMs performed near random on weed identification and open-source models lag behind proprietary services [21]. Comparisons of VLM datasets indicate that Agri-LLaVA uses synthetic instructions, while AgroBench provides real images with expert annotations [21]. AgriCoT introduced a 4 535-sample chain-of-thought dataset and revealed that existing models struggle with reasoning [22]. Another multi-dataset study reported that even the best proprietary VLM achieves only 62% accuracy, with open-ended tasks below 25% [23]. Multi-modal reasoning for agriculture remains an open challenge.

III. Applications: Detection, Segmentation, Classification and Prediction

Deep learning enables diverse tasks in agriculture, ranging from visual diagnosis to yield forecasting. This section organizes applications by problem type.

A. Disease and Pest Detection

Plant disease detection is one of the earliest and most studied DL applications. CNNs classify leaf images for diseases such as late blight, powdery mildew and rust. Mohanty et al. demonstrated that AlexNet and GoogleNet achieved over 99% accuracy on PlantVillage images [9], but performance drops under field conditions. To improve generalization, researchers combined datasets and used noise augmentation and class weighting; a 2025 study fine-tuned EfficientNet on PlantDoc and web images, achieving 80.2% cross-dataset accuracy and F1 scores above 90% for several disease classes [24]. Depthwise CNNs with squeeze–excitation and residual connections reached 98% accuracy and F1 = 98.2% on plant disease datasets [25]. Edge-oriented pest detection models fused MobileNetV2 and EfficientNet to achieve high precision and F1 scores [10]. Generative adversarial networks augment scarce disease datasets and improve robustness [5], [26].

B. Weed Detection and Segmentation

Weeds reduce yields and increase herbicide use. Early studies segmented vegetation using color indices before classification. An unsupervised CNN weed detector automatically identified crop rows and extracted inter-row weeds as training data; its accuracy was within 1.5–6% of a supervised baseline [27]. Subsequent research focused on real-time detection with YOLO. A systematic review of 61 CNN weed studies found that YOLO variants dominate and that digital cameras, smartphones and UAVs are primary data sources [14]. CNNs require large labeled datasets and computing resources [14]. HDMS-YOLO improved detection accuracy by incorporating hierarchical features, while YOLO-Weed-Nano achieved similar or better mAP with drastically fewer parameters [15]. WeedNet-ViT and WeedSwin use transformers and SAM to segment weeds accurately [17], [20]. U-Net variants with attention modules deliver high MIoU and F1 scores [3], [12], and semi-supervised versions reduce annotation costs. Weed segmentation remains challenging due to class imbalance and spectral variability.

C. Stress Monitoring and Yield Prediction

DL models predict drought stress, soil moisture and crop yield by integrating remote sensing, meteorological and management data. Hyperspectral stress detection using new indices and a 1D CNN detected stress 10-15 days earlier than conventional indices and achieved 83.40% accuracy [28]. Soil moisture prediction with a hybrid LSTM-XGBoost model reached $R^2 \approx 0.985$ for 24-168 hour forecasts [8]. A CNN-RNN yield predictor achieved RMSE around 9% of average yield and generalized across regions [2]. Multi-temporal models using Landsat images across 11 years and pixel augmentation produced overall accuracy 89.44% for wheat, soybean and corn yields [29]. A CNNAtBiGRU model combining 1D CNN, bidirectional GRU and attention achieved $R^2 = 0.896$ and RMSE 908.33 $kg.ha^{-1}$ for maize and provided predictions one to two months before harvest [30]. Long-term yield studies indicate that CNNs and LSTMs are dominant models [31], but dataset heterogeneity hinders general comparisons. Future work aims to integrate genotype, management and socio-economic data for more holistic forecasts.

D. Farm Management and Robotics

DL supports autonomous machines for harvesting, spraying and navigation. Multi-agent reinforcement learning (RL) coordinates dual-arm apple-picking robots; a cooperative strategy combining Multi-Agent Proximal Policy Optimization and a greedy picking algorithm reduced path length by 15.1% and achieved 92.3% success rate [6]. Re-DQN coverage path planning reduced path length by 31.56% and navigation time by

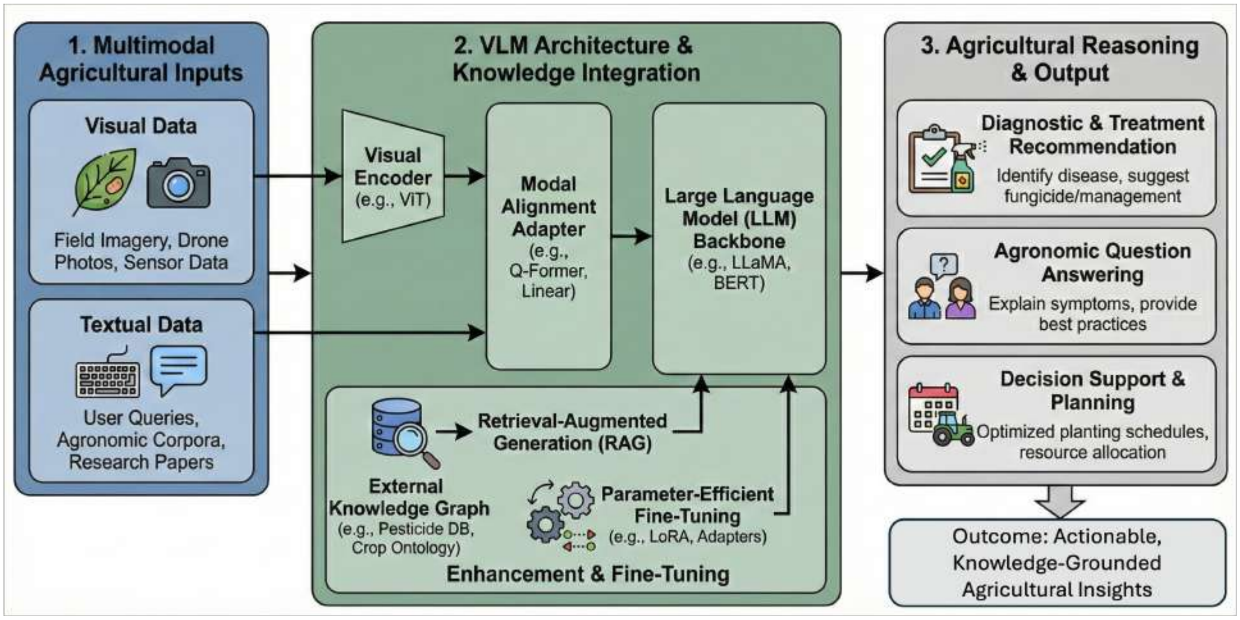

Fig. 4: Overview of transformer model and its application in agriculture.

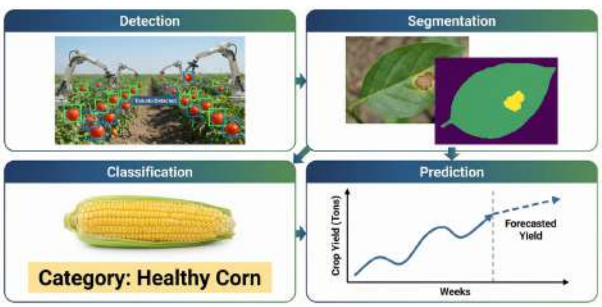

Fig. 5: Applications of deep learning in agriculture (detection, segmentation, classification and prediction).

35.72% compared to a boustrophedon baseline when navigating kiwifruit orchards [32]. Lightweight CNNs and mixtures of small models enable real-time perception on agricultural robots [12]. However, simulation environments often oversimplify dynamics, causing discrepancies between simulated and real orchards [6]. Integrating perception, planning and control into end-to-end models remains an open research area.

E. Vision–Language Reasoning

VLMs attempt to answer multi-modal questions about crop images and metadata. AgroBench revealed that zero-shot VLMs perform near random on weed identification and general crop classification [21]. The AgriCoT dataset showed that chain-of-thought reasoning is weak in current models, while a multi-dataset benchmark found that even the best proprietary VLM achieved only 62% accuracy [22]. To improve multi-modal reasoning, models need domain-specific pre-training and retrieval-augmented generation, as illustrated by AgriGPT [7]. Combining vision and language features holds promise for comprehensive farm advisory systems but remains at an early stage.

IV. Mobile and Web Platforms

Delivering DL services to farmers requires mobile and web platforms that operate under connectivity and resource constraints. Edge computing frameworks compress models and run them locally, reducing latency and preserving privacy [13]. The Plantix system integrates sensors and a mobile app to detect plant diseases and provides real-time diagnosis and treatment recommendations [13]. The Cocoa Companion app deploys an SSD MobileNet V2 model to diagnose cocoa diseases; it achieved more than 80% confidence for Swollen Shoot and Black Pod diseases in field images [33]. Another cross-platform mobile application trained a custom CNN on PlantVillage and achieved 92.06% test accuracy, integrating severity estimation and providing real-time feedback via a React Native interface [34]. These apps demonstrate the feasibility of delivering DL through smartphones, but reliability under varied lighting and network conditions remains a challenge. Recent work explores on-device weed detection and yield prediction using lightweight models such as YOLO-Weed-Nano and small CNNs [16]. Web-based platforms allow integration of remote sensing data and farm management dashboards, yet their adoption depends on broadband access and digital literacy in rural areas.

V. Fine-Tuning for the Agricultural Domain

Fine-tuning adapts pre-trained networks to domain-specific tasks. In weed detection, transfer learning and fine-tuning on the DeepWeeds dataset yielded efficient models; EfficientNetV2B1 achieved 94.2% accuracy, while InceptionV3, VGG16 and DenseNet121 offered similar accuracy at higher training cost [35]. Fine-grained weed recognition with Swin Transformer used a contrastive loss and a two-stage transfer strategy: the network was first pre-trained on ImageNet, fine-tuned on the Plant Seedlings dataset, and then fine-tuned on a private maize–weed dataset (MWFI), resulting in approximately 99% accuracy, precision, recall and F1-score [36]. These examples show that freezing early layers and gradually fine-tuning deeper layers on increasingly specific datasets improves performance with limited data.

Domain-specific LLMs and VLMs also leverage fine-tuning. AgriGPT synthesized an expert-tuning dataset by generating natural language descriptions from six image-only datasets covering diseases, weeds, insects and fruits; it then created question–answer pairs and performed a two-stage instruction tuning followed by expert tuning, yielding a domain-specialized model that outperformed general LLMs [37]. AgriBERT was pre-trained from scratch on a 300-million-token agricultural corpus and fine-tuned with knowledge graph–guided augmentation, which improved semantic matching tasks in food and nutrition domains [38]. A multimodal model for crop disease and pest identification used a Q-Former to align visual and textual features and applied low-rank adaptation (LoRA) to adjust the weights of a pre-trained language model, enabling accurate recognition and professional recommendations [39]. Reviews note that fine-tuning large models can be feature-based (extract features with frozen backbones) or full fine-tuning and that transformers form the backbone of modern LLMs and VLMs. Generative multimodal LLMs can be fine-tuned on agricultural data to generate synthetic training examples and augment small datasets [40].

Fine-tuning strategies vary. Standard transfer learning freezes low-level layers trained on large datasets and adjusts higher layers on agricultural images or text. Two-stage transfer learning pre-trains on an intermediate domain (e.g., Plant Seedlings) before fine-tuning on a target dataset, as demonstrated in weed recognition [36]. Low-rank adaptation inserts trainable rank-decomposed matrices into linear layers to reduce the number of trainable parameters [39]. Retrieval-augmented generation pairs an LLM with a search engine to supply relevant agronomic facts during fine-tuning [37]. Knowledge infusion incorporates external graphs into fine-tuning to embed agronomic concepts [38]. When datasets are small, self-supervised pre-training and domain-specific pre-training supply better initial weights before fine-tuning [41].

Rules of thumb for fine-tuning include matching the depth of fine-tuning to data availability: when only a small dataset is available, freeze the majority of convolutional or transformer layers and train a lightweight classifier head; as more data become available, progressively unfreeze deeper layers and reduce the learning rate to avoid catastrophic forgetting. Data augmentation and regularization (e.g., dropout, weight decay) are essential to prevent overfitting. Domain-specific normalization and balancing are important because agricultural datasets often suffer from class imbalance and distribution shift. For sequence models and transformers, gradient clipping and learning rate warm-up help stabilize training. In language and vision–language models, low-rank adaptation (LoRA) and prompt-tuning offer efficient alternatives to full fine-tuning, reducing memory requirements. Retrieval-augmented generation adds external knowledge at inference time and alleviates the need for large fine-tuning corpora, while knowledge graph infusion ensures models learn explicit agronomic relationships. These strategies allow practitioners to adapt pre-trained models to diverse agricultural tasks under resource constraints.

VI. Performance Metrics

Evaluating deep models requires appropriate metrics. Precision quantifies the proportion of correctly predicted positives among all positive predictions, recall measures the proportion of actual positives that are correctly identified, and the F1-score is the harmonic mean of precision and recall. Intersection over Union (IoU) quantifies the overlap between predicted and ground-truth bounding boxes or masks. Mean Average Precision (mAP) averages the precision–recall area across classes and is the standard for object detection [42]. Pixel Accuracy (PA) is the ratio of correctly classified pixels to all pixels, mean Pixel Accuracy (mPA) averages PA across classes, and Mean IoU (MIoU) averages IoU across classes [43]. Regression tasks use metrics such as mean absolute error (MAE), mean squared error (MSE), root mean squared error (RMSE), coefficient of determination (R^2) and mean absolute percentage error (MAPE) to measure discrepancies between predicted and observed values [44]. Computational efficiency is reported in giga floating-point operations (GFLOPs) for detection models [42]. Table I summarises commonly used metrics and their typical applications.

Evaluating performance requires understanding the strengths and weaknesses of each metric. Accuracy is simple but can be misleading when classes are imbalanced; F1 balances precision and recall yet ignores true negatives. IoU and MIoU quantify spatial overlap but penalize small objects, while pixel accuracy can appear high even when minority classes are misclassified. Regression metrics such as RMSE emphasize larger errors and are sensitive to outliers. GFLOPs measure computational demand but not runtime on specific hardware. For language models, BLEU and ROUGE rely on n-gram overlap and may not capture semantic equivalence [47]. Perplexity gauges how well a model predicts tokens but applies only to autoregressive models [49], [51]. ANLS provides soft matching that tolerates minor spelling variations and is useful for evaluating VLMs on OCR and VQA tasks [50], [52], [53]. Careful metric selection aligned with the application and dataset characteristics is therefore critical.

VII. Benchmark Datasets

Comparative evaluation of DL models requires standardized datasets. Table II summarizes major datasets used in agricultural DL research.

TABLE I: Common performance metrics used in agricultural machine learning, including vision, regression, and language and vision-language evaluation.

Metric	Definition	Application or Task
Precision	Ratio of true positives to all predicted positives [42]	Classification, detection
Recall	Ratio of true positives to all actual positives [45]	Classification, detection
F1-score	Harmonic mean of precision and recall [42]	Classification, detection, segmentation
Accuracy	Proportion of correctly predicted samples among all samples [46]	Classification
Intersection over Union (IoU)	Area of overlap divided by area of union between prediction and ground truth [42]	Detection, segmentation
Mean Average Precision (mAP)	Mean of average precision across classes [42]	Object detection
Pixel Accuracy (PA)	Correctly classified pixels divided by total pixels [43]	Segmentation
Mean Pixel Accuracy (mPA)	Average of PA values over classes [43]	Segmentation
Mean IoU (MIoU)	Average IoU across all classes [43]	Segmentation
Mean Absolute Error (MAE)	Average absolute difference between predicted and true values [44]	Regression, yield and moisture
Mean Squared Error (MSE)	Average squared difference between predicted and true values [44]	Regression
Root Mean Squared Error (RMSE)	Square root of MSE, indicating residual standard deviation [44]	Regression
Coefficient of Determination (R^2)	Fraction of variance in observations explained by the model [44]	Regression
Mean Absolute Percentage Error (MAPE)	Mean of absolute percentage errors [44]	Regression
GFLOPs	Billions of floating-point operations used during inference [42]	Computational cost
BLEU	N-gram precision metric comparing generated text with one or more reference translations [47]	Machine translation, text generation, summarization in LLMs
ROUGE	N-gram recall metric measuring overlap between generated and reference sequence [48]	Summarization and translation tasks in LLMs
Perplexity	Exponentiated average negative log-likelihood of a sequence, lower values indicate better predictive ability [49]	Generative language modelling
ANLS	Average normalized Levenshtein similarity, measures character-level similarity between prediction and ground truth [50]	Visual question answering and optical character recognition in vision–language models

VIII. Comparison Table

Table III compares representative studies across key applications. It lists the year, citation, methodology or approach, important variables or dataset characteristics, the main finding and a noted limitation. The table highlights how performance depends on data quality, architectural choices and task complexity.

IX. Deep Learning Techniques and Open Directions

A. Data Augmentation and Generative Methods

DL models often overfit due to small or imbalanced datasets. Researchers apply geometric transformations, color jittering, and noise injection to augment images. GANs generate realistic synthetic samples for rare disease classes [5]. Domain-specific augmentations combine spectral indices or environmental metadata. Self-supervised learning frameworks such as SimCLR pre-train encoders on unlabeled agricultural images, reducing annotation costs and improving downstream performance [41]. Augmentation should reflect real variations in lighting, growth stage and background.

B. Domain Adaptation and Transfer Learning

Domain shift—differences in sensor, location or crop variety—degrades model performance. A review of domain adaptation methods in agricultural image analysis categorizes approaches into supervised, semi-supervised and unsupervised strategies based on adversarial learning and feature alignment. Cross-region weed detection using adversarial domain adaptation improved accuracy by aligning feature distributions between regions [57]. Transfer learning from ImageNet or COCO remains common; however, domain-specific pre-training on large agricultural datasets may yield better representations for weeds and diseases.

C. Semi-Supervised and Self-Supervised Learning

Annotation of agricultural images is costly. Semi-supervised models like SemiWeedNet use a small labeled set and a larger unlabeled set with consistency regularization [12]. Self-supervised learning frameworks pre-train a backbone using contrastive objectives, then fine-tune on limited labeled data [41]. These methods reduce the need for human annotation and improve generalization across environments. Cross-attention gating (CAG) modules and dynamic attention gating (DAG) have been incorporated into segmentation networks to focus on relevant spectral bands and spatial regions

TABLE II: Representative datasets used in agricultural machine learning, spanning plant disease, pest and weed analysis, and language and vision–language tasks.

Dataset	Task	Classes or Images	Notes
PlantVillage [13]	Disease classification	$>$ 54,000 images, 38 crop–disease classes	Controlled conditions, widely used but lacks field variability
PlantDoc [24]	Disease classification	2,590 images, 13 disease classes	Field images, combined with web images for generalization
Rice Leaf Diseases [13]	Disease classification	433 images, 3 classes	Local dataset for leaf diseases
PDDB [13]	Disease classification	13,329 images, 32 classes	Field images with environmental variability
New Plant Diseases [13]	Disease classification	2,560 images, 8 classes	Augmented dataset with rotated and scaled images
IP102 [13]	Insect pest classification	75,222 images, 102 insect species	Large pest dataset with imbalanced classes
DeepWeeds [20]	Weed classification	17,509 images, 8 weed species	Field images with varying backgrounds, used for WeedNet-ViT
CWFID [3]	Weed segmentation	3 datasets combined, approx. 28,000 images	Used to evaluate attention U-Net variants
WeedSwin [17]	Weed detection and segmentation	16 weed species, 11 growth stages	Large dataset supporting transformer training
Cotton Weed [16]	Weed detection	27,785 images, 4 weed classes	Used to train YOLO-Weed-Nano
CCMT Dataset [10]	Pest and disease detection	Unspecified images, multiple species	Used for MobileNetV2–EfficientNetB0 fusion model
AgriBench-13K [7]	LLM QA	13k QA pairs	Part of AgriGPT, covered multi-modal instructions
AgroBench [21]	Vision–language benchmark	203 crops, 682 diseases, real images	Expert annotated, measured zero-shot performance
AgriCoT [22]	Vision–language reasoning	4,535 samples	Chain-of-thought prompts testing reasoning

[12]. Domain-specific pre-training (DSP) uses large unlabeled agricultural datasets for pretraining, which may be combined with fine-grained feature alignment in later stages.

D. Interpretable and Uncertainty-Aware Models

Adoption of DL in agronomy requires interpretability. Attention maps and saliency measures help confirm that models focus on disease spots or weed patches [20]. Monte Carlo dropout and Bayesian neural networks estimate prediction uncertainty, guiding decisions such as when to consult experts. Incorporating explainability improves trust and supports diagnostic decisions. Research also explores chain-of-thought prompting in VLMs to improve reasoning [22].

E. Project Ideas and Future Directions

- **Multimodal Transformer for Crop Health Monitoring:** Build a transformer that fuses hyperspectral, RGB, thermal and environmental time-series data. Hypothesis: self-attention across modalities will improve early stress detection by at least 5% over current models.
- **Federated Learning for Yield Prediction:** Train yield prediction models across farms without exchanging raw data. Compare federated models to centralized baselines to evaluate performance and privacy benefits.
- **Edge AI Weed Sprayer:** Develop a lightweight detection and segmentation model (e.g., YOLO-Weed-Nano) deployed on a sprayer to identify weeds and control nozzles in real time. Aim to achieve $> 90\%$ precision and significant herbicide reduction.
- **AgriChat: Vision-Language Assistant:** Create a VLM fine-tuned on AgroBench, AgriBench-13K and AgriCoT to provide diagnostic support and farm management recommendations. Evaluate chain-of-thought reasoning and domain retrieval using open-source LLMs.
- **Robotic Harvesting via Multi-Agent RL:** Extend MAPPO and re-DQN frameworks to coordinate multiple ground and aerial robots for fruit harvesting and yield mapping. Test in realistic orchards to achieve picking success above 90%.

X. Conclusion

This review examined deep learning research in agriculture since 2010, organizing studies by architecture, application, platform, dataset, evaluation metric and fine-tuning strategy. Convolutional networks have dominated disease and weed classification, but generalization beyond controlled conditions remains limited. Attention-enhanced U-Nets and transformer models such as WeedSwin achieve high segmentation and detection accuracy yet require significant computational resources and large datasets. Overall, deep learning has transformed precision agriculture by enabling automated perception and decision support. Research should now focus on collecting diverse field-level datasets, developing multimodal models that fuse spectral, spatial and textual information, and integrating uncertainty estimation and interpretability. Future work should also address cross-region domain adaptation, federated learning and responsible deployment that considers data privacy and farmer trust. Deep learning is poised to play an increasingly important role in sustainable agriculture,

TABLE III: Representative methodologies in agricultural machine learning, with key settings, headline results, and primary limitations.

Methodology	Key parameters	Main finding	Critical limitation
AlexNet and GoogleNet [54]	PlantVillage dataset, $>$ 54k leaf images, 38 classes	Achieved $>$ 99% accuracy for diseases under controlled conditions	Models exhibited poor generalization to field conditions and unseen species
WeedSwin with SAM-2 [4]	Hierarchical ViT, self-attention and SAM-2 modules, 16 weed species	mAP 0.993 and mean average recall 0.985 at 218 fps	Requires large datasets and high computational resources; limited crop evaluation
Attention U-Net with CBAM [55]	Combined CWFID, Sugar Beet and Sunflower datasets, 28k images	99.09% accuracy, mean IoU 81.02% and F1-score 99.06%	Training and validation limited to specific crops; class imbalance challenges
EDM-UNet [12]	UAV weed segmentation, ECA module, Canny edge detection	Mean IoU 89.45% and real-time inference at 40 fps	Relied on post-processing; performance may drop in complex backgrounds
CNN–RNN Hybrid [56]	Weather, soil and management data for corn/soybean	RMSE 8–9% of average yields; generalized across regions	Lacked genotype data; may struggle to extrapolate extreme climatic conditions
Hybrid LSTM–XGBoost [8]	Sensor time series, LSTM temporal and XGBoost non-linear modules	$R^2 \approx 0.985$ for 24- to 168-hour forecasts	Data from limited regions; requires dense sensor deployment
1D CNN + MLVI [28]	Hyperspectral images, MLVI/H-VSI indices, recursive feature elimination	83.40% accuracy; detected stress 10–15 days earlier than conventional indices	Hyperspectral data collection is costly and may not scale across farms
Deep Regression [29]	Landsat imagery (11 seasons), 2×2 pixel augmentation	89.44% overall accuracy across wheat, soybean, and corn	Depends on cloud-free satellite images; limited to large-scale fields
YOLO-Weed-Nano [16]	Cotton weed dataset, depthwise separable convolutions, BiFPN	Parameters reduced 63.8%; mAP improved by 1% over YOLOv8n	Limited weed classes; requires adaptation for other crops/species
MobileNetV2–EffNet [10]	CCMT dataset, quantization and pruning for edge deployment	89.5% accuracy, 95.68% precision; runs on low-power devices	Limited pest/disease coverage; generalization to various lighting uncertain
Transfer Learning [35]	DeepWeeds dataset, EfficientNetV2B1, InceptionV3, VGG16	EfficientNetV2B1 achieved 94.2% accuracy	Reliance on pre-trained models may limit adaptation to new weed species
Two-stage Swin [36]	Pre-trained ImageNet, fine-tuned on Plant Seedlings and MWFI	$\approx$ 99% accuracy and F1-score for fine-grained recognition	Small private dataset restricted generalization; resource-intensive training
AgriGPT [7]	70k instruction-tuning dataset, AgriBench-13K QA corpus	Specialized LLM outperformed general models on agricultural QA	Limited to text; requires updates as agronomic knowledge evolves
AgroBench VLMs [21], [23]	203 crop classes, 682 disease categories, multiple VLMs	Zero-shot VLMs performed near random; proprietary models reached 62%	Lack of agricultural pre-training; struggled with open-ended reasoning

provided that researchers continue to address data, model and deployment challenges.

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